

EEGPT Unified60 Equivalence Detailed Report

Original reproduction vs unified60 ablation - seed 42, 15 epochs, matched subset. Generated in a consistent format for GitHub tracking.

Verdict: EEGPT is resolved for matched-subset equivalence tracking by identifying checkpoint mapping/loading behavior and original-recipe DataLoader settings as the key factors. The strongest reproduction-matched unified60 row is A3_original_recipe_plus_checkpoint_none: B-Acc 0.7042 vs original 0.7140 (-0.98 pp), and AUROC 0.8267 vs original 0.7948 (+3.19 pp). Guardrail: mapping OFF is a diagnostic original-behavior control, not a universal scientific recommendation.

1. Scope and comparison rule

This report separates three evidence levels: paper TUAB reference, local original reproduction on the matched subset, and controlled unified60 H5 ablations. The paper row is a full-dataset sanity anchor only; the fair comparison is original reproduction vs matched-subset unified60 rows.

Unlike the bipolar-chain models, EEGPT already uses the same [B,23,2000] input shape in original and unified60. Therefore, the ablation focuses on optimizer/DataLoader recipe and checkpoint mapping behavior rather than montage reconstruction.

2. Main original reproduction vs unified60 pipeline

Stage	Original reproduction pipeline	Unified60 A3 reproduction-matched pipeline
Data source	Original PKL matched subset	Canonical unified60 H5 matched subset
Input shape	[B,23,2000]	[B,23,2000]
Pretrained behavior	Original logs suggested encoder.* keys may not have been fully used	Checkpoint mapping disabled (0 keys loaded) to mimic original-observed behavior
Optimizer/DataLoader	lr 5e-4, wd 0.05, workers 2, train drop_last True	same original-style recipe
Evaluation target	Original subset reference: B-Acc 0.7140, AUROC 0.7948	A3: B-Acc 0.7042, AUROC 0.8267

3. Reference anchors and ablation design

Row	Data source	Mapping	Recipe	Question answered
Original reproduction	Original PKL	Observed/unclear	lr 5e-4, wd 0.05, workers 2, drop_last True	Subset target baseline
A0 current unified	Unified60 H5	ON, 102 keys	lr 1e-4, wd 0.0005, workers 8, drop_last False	Does launcher reproduce previous unified60?
A1 original recipe	Unified60 H5	ON, 102 keys	original optimizer/DataLoader	Does recipe alone close the gap?
A2 mapping none	Unified60 H5	OFF, 0 keys	current unified recipe	Does checkpoint mapping drive the gap?
A3 recipe + mapping none	Unified60 H5	OFF, 0 keys	original optimizer/DataLoader	Do both original-like behaviors move unified60 closest to original?

4. Ablation results and gap against original

Row	Best epoch	B-Acc	AUROC	B-Acc gap	AUROC gap	Interpretation
Original reproduction	13	0.7140	0.7948	-	-	reference
Previous unified / A0	2	0.6906	0.7729	-2.34 pp	-2.20 pp	old mismatch baseline
A1 original recipe	6	0.6588	0.7371	-5.52 pp	-5.77 pp	recipe alone worsened

Row	Best epoch	B-Acc	AUROC	B-Acc gap	AUROC gap	Interpretation
A2 mapping none	5	0.7028	0.7846	-1.12 pp	-1.02 pp	mapping behavior matters
A3 recipe + mapping none	14	0.7042	0.8267	-0.98 pp	+3.19 pp	strongest reproduction-matched row

5. Findings, settings, and guardrails

- A0 exactly reproduced the previous unified60 baseline, validating the ablation launcher.
- Original optimizer/DataLoader settings alone did not close the gap.
- Disabling checkpoint mapping/loading was the strongest tested factor; A3 also aligned the best epoch with the original run.
- A3 should be described as a reproduction-behavior control because original checkpoint loading behavior was ambiguous.
- Recommended final-row wording: EEGPT unified60 is resolved for tracking using A3 as the strongest reproduction-matched row; A0 remains the strict current-unified baseline.

6. Reproducibility notes

Item	Value
Source report	eegpt_unified60_ablation_seed42_report.pdf
Output root	outputs/eegpt_unified60_ablation_seed42_2gpu
Main unified row	A3_original_recipe_plus_checkpoint_none
Reference original row	original_reproduction_reference
Epochs / seed	15 epochs / seed 42

BIOT Unified60 Equivalence Detailed Report

Original reproduction vs patched unified60 ablation - seed 42, 15 epochs, matched subset. Generated in a consistent format for GitHub tracking.

Verdict: BIOT is equivalent after using the author-style scratch configuration. The fair row is B1_bipolar_scratch_recipe: true bipolar reconstruction, q95 normalization, BCE loss, Adam lr=1e-3/wd=1e-5, no PREST checkpoint, FFT/token size 200, hop 100, and best validation AUROC. B1 reaches B-Acc 0.7646 and AUROC 0.8449 vs original B-Acc 0.7627 and AUROC 0.8512.

1. Scope and comparison rule

The fair comparison is the matched original BIOT-style PKL reproduction against unified60 H5 rows using the same intended scratch recipe. PREST-pretrained rows are kept as useful controls but are not the fair scratch baseline.

2. Main original reproduction vs unified60 pipeline

Stage	Original BIOT reproduction	Debugged unified60 B1
Data source	Original BIOT-style PKL from matched EDF subset	Canonical unified60 H5 matched subset
Input / montage	True 16-channel bipolar, 10 s at 200 Hz, shape (16,2000)	true bipolar_qnorm adapter, [batch,16,2000]
Normalization	q95 absolute normalization in loader	q95 absolute normalization in adapter/loader
Model / loss	BIOTClassifier / BCEWithLogitsLoss	BIOTClassifier / BCEWithLogitsLoss
Optimizer / checkpoint	Adam lr=0.001, wd=1e-5, checkpoint OFF	same scratch recipe, checkpoint OFF
Evaluation	best val AUROC, B-Acc 0.7627 / AUROC 0.8512	best val AUROC, B-Acc 0.7646 / AUROC 0.8449

3. Ablation design

Row	Adapter	Recipe	Checkpoint	Question answered
Previous unified60	endpoint/current	old unified	PREST/checkpointed + CE mismatch	Reproduces old non-equivalent baseline
B0 endpoint scratch	current	github_vanilla	OFF	Does scratch recipe work with old endpoint adapter?
B1 bipolar scratch	bipolar_qnorm	github_vanilla	OFF	Fair scratch row: does true bipolar + q95 match original?
B2 endpoint pretrained	current	github_pretrained	ON	PREST effect with old adapter
B3 bipolar pretrained	bipolar_qnorm	github_pretrained	ON	PREST control under corrected bipolar input

4. Ablation results

Row	Best epoch	B-Acc	Delta B-Acc	AUROC	Delta AUROC	Status
Original reproduction	5	0.7627	-	0.8512	-	reference
Previous unified60	9	0.7536	-0.91 pp	0.8202	-3.10 pp	not equivalent
B0 endpoint scratch	15	0.6960	-6.67 pp	0.8343	-1.70 pp	wrong adapter

Row	Best epoch	B-Acc	Delta B-Acc	AUROC	Delta AUROC	Status
B1 bipolar scratch	6	0.7646	+0.19 pp	0.8449	-0.63 pp	main equivalent row
B2 endpoint pretrained	11	0.7653	+0.26 pp	0.8488	-0.24 pp	control
B3 bipolar pretrained	6	0.7840	+2.13 pp	0.8527	+0.15 pp	best but not fair scratch

5. Findings and settings

- The previous unified60 row mixed PREST loading, CrossEntropyLoss, and endpoint channel selection, so it was not a fair scratch reference.
- True bipolar reconstruction was the key fix; B1 recovered original-level balanced accuracy.
- B3 shows PREST can improve performance, but it answers a different question because the original fair reference is scratch.
- Recommended final-row wording: BIOT is equivalent after patch; use B1 as the fair unified60 row and keep B3 as a pretrained control.

6. Reproducibility notes

Item	Value
Source report	biot_unified60_ablation_seed42_report.pdf
Output root	outputs/biot_unified60_ablation_seed42_1gpu
Fair row metrics path	outputs/biot_unified60_ablation_seed42_1gpu/B1_bipolar_scratch_recipe/metrics.json
Original metrics path	outputs/all6_original_repo_vs_90job_unified60_matchedsplit_epoch15_seed42_v1/BIOT/original/metrics.json
Previous unified path	outputs/all6_unified60_subset_epoch15_5seed_1gpu_dev_v1/biot/full_finetune/seed_42/metrics.json

CBraMod Unified60 Equivalence Detailed Report

Original reproduction vs patched unified60 ablation - seed 42, 15 epochs, matched subset. Generated in a consistent format for GitHub tracking.

Verdict: CBraMod is equivalent after correcting the input adapter and checkpoint mapping. The fair row is C3_adapter_plus_author_recipe: H5 bipolar_div100 input, author recipe, checkpoint mapping ON with 209 keys, and all_patch_reps model. C3 reaches B-Acc 0.7896 and AUROC 0.8693 vs original B-Acc 0.7891 and AUROC 0.8544.

1. Scope and comparison rule

The paper TUAB row is a sanity reference only. The fair comparison is original reproduction vs matched unified60 variants on the 8192/2048/2048 subset. The original CBraMod reproduction is treated as valid, so the ablation isolates adapter, recipe, and checkpoint mapping effects.

2. Main original reproduction vs unified60 pipeline

Stage	Original CBraMod reproduction	Debugged unified60 C3
Data source	Original matched PKL subset	Canonical H5 unified60 matched subset
Input construction	16 bipolar derivations with /100 scaling	bipolar_div100 reconstructs same bipolar derivations and divides by 100
Model	CBraMod all_patch_reps, 69.3M params	same model size restored
Checkpoint	author checkpoint loaded	backbone_prefix mapping loads 209 keys
Recipe	author recipe: WD 0.05, cosine per batch, grad clip 1.0	same author recipe
Evaluation	Original B-Acc 0.7891 / AUROC 0.8544	C3 B-Acc 0.7896 / AUROC 0.8693

3. Ablation design

Row	Adapter	Recipe	Checkpoint	Question answered
C0 current unified	current endpoint	unified wd 0.0005, no scheduler	ON 209 keys	Reproduced old/current unified behavior
C1 adapter only	bipolar_div100	unified	ON 209 keys	Does adapter alone solve most of the gap?
C2 author recipe only	current endpoint	author wd 0.05 + cosine	ON 209 keys	Does recipe alone solve the gap?
C3 adapter + author recipe	bipolar_div100	author	ON 209 keys	Final fair row
C4 checkpoint OFF	bipolar_div100	author	OFF, 0 keys	Checkpoint contribution control

4. Ablation results and effects

Row	Best epoch	B-Acc	AUROC	Adapter	Recipe	Decision
Original reproduction	5	0.7891	0.8544	PKL bipolar	author	target baseline
Previous unified60	3	0.7470	0.8189	endpoint	unified	non-equivalent
C0 current unified	3	0.7538	0.8401	current	unified	baseline
C1 adapter only	4	0.8027	0.8720	bipolar_div100	unified	adapter helps most

Row	Best epoch	B-Acc	AUROC	Adapter	Recipe	Decision
C2 author recipe only	2	0.7260	0.8100	current	author	recipe alone fails
C3 patched equivalent	5	0.7896	0.8693	bipolar_div100	author	main equivalent row
C4 checkpoint OFF	6	0.7253	0.7984	bipolar_div100	author	checkpoint matters

5. Findings and settings

- The dominant blocker was the endpoint-channel adapter; author-equivalent CBraMod expects 16 bipolar derivations and /100 scaling.
- C1 showed the adapter effect alone produced the largest positive shift.
- C2 proved the author recipe alone does not fix a wrong input.
- C3 matched original B-Acc almost exactly and improved AUROC.
- C4 confirmed compatible pretrained checkpoint loading is important.

6. Reproducibility notes

Item	Value
Source report	cbramod_unified60_ablation_seed42_report.pdf
Output root	outputs/cbramod_unified60_ablation_seed42_2gpu
Fair row	C3_adapter_plus_author_recipe
Main metrics	C3 B-Acc 0.789587 / AUROC 0.869287
Epochs / seed / batch	15 epochs / seed 42 / batch size 64

CodeBrain Unified60 Equivalence Detailed Report

Original reproduction vs patched unified60 ablation - seed 42, 15 epochs, matched subset. Generated in a consistent format for GitHub tracking.

Verdict: CodeBrain is equivalent after using the reproduction wrapper recipe and bipolar_div100 adapter. The fair row is C1_bipolar_original_recipe: B-Acc 0.7897 and AUROC 0.8490 vs original B-Acc 0.7653 and AUROC 0.8510. This is +2.44 pp B-Acc and -0.19 pp AUROC relative to original.

1. Scope and comparison rule

The inspected CodeBrain repo required wrapper plumbing for a runnable matched-subset TUAB comparison. Therefore, the fair comparison is not a random author_default run, but the same original_wrapper logic applied to canonical unified60 H5 input.

The raw/default TUAB forward was not treated as the fair target because the inspected path was not directly compatible with the matched-subset tensor shape.

2. Main original reproduction vs unified60 pipeline

Stage	Original CodeBrain reproduction	Debugged unified60 C1
Data source	Original branch on Option 1 matched EDF subset	Canonical unified60 H5 matched subset
Adapter/input	Wrapper-compatible CodeBrain TUAB input	bipolar_div100 true bipolar reconstruction
Forward path	Wrapper-only flattened TUAB forward; broken raw TUAB path not used	adapter-side flattened TUAB forward, same idea
Recipe	original_wrapper	original_wrapper
Checkpoint	ON through wrapper	ON, 269 loaded keys
Evaluation	Original B-Acc 0.7653 / AUROC 0.8510	C1 B-Acc 0.7897 / AUROC 0.8490

3. Ablation design

Row	Adapter	Recipe	Checkpoint	Question answered
C0 endpoint original recipe	current endpoint	original_wrapper	ON 269 keys	Baseline with old adapter under fair wrapper recipe
C1 bipolar original recipe	bipolar_div100	original_wrapper	ON 269 keys	Main fair row
C2 endpoint author recipe	current endpoint	author_default	ON 269 keys	Does author recipe help old adapter?
C3 bipolar author recipe	bipolar_div100	author_default	ON 269 keys	Author-default control with corrected adapter
C4 checkpoint off	bipolar_div100	author_default	OFF	Checkpoint contribution control

4. Ablation results

Row	Best epoch	B-Acc	Delta B-Acc	AUROC	Delta AUROC	Status
Original reproduction	2	0.7653	-	0.8510	-	reference
Previous unified60	9	0.7437	-2.16 pp	0.8053	-4.57 pp	not equivalent
C0 endpoint original recipe	7	0.7539	-1.15 pp	0.8309	-2.01 pp	adapter issue

Row	Best epoch	B-Acc	Delta B-Acc	AUROC	Delta AUROC	Status
C1 bipolar original recipe	3	0.7897	+2.44 pp	0.8490	-0.19 pp	main equivalent row
C2 endpoint author recipe	4	0.7594	-0.59 pp	0.8266	-2.44 pp	control
C3 bipolar author recipe	3	0.7759	+1.05 pp	0.8422	-0.87 pp	useful control
C4 checkpoint off	3	0.7367	-2.86 pp	0.8034	-4.76 pp	control only

5. Findings and settings

- C1 is the fair reproduction-equivalent row because it keeps the original_wrapper recipe and only replaces the data source with canonical H5 via bipolar_div100.
- C3 answers a different question: author_default recipe with bipolar_div100. It is useful but not the exact fair row.
- Checkpoint loading matters under the author-default control, as C3 substantially exceeds C4.
- Recommended final-row wording: CodeBrain unified60 is equivalent after adapter/wrapper patch; use C1 in the cross-model table.

6. Reproducibility notes

Item	Value
Source report	codebrain_unified60_ablation_seed42_report.pdf
Output root	outputs/codebrain_unified60_ablation_seed42_2gpu
Fair row metrics path	outputs/codebrain_unified60_ablation_seed42_2gpu/C1_bipolar_original_recipe/metrics.json
Original metrics path	outputs/all6_original_repo_vs_90job_unified60_matchedsplit_epoch15_seed42_v1/CodeBrain/original/metrics.json
Previous unified metrics path	outputs/all6_unified60_subset_epoch15_5seed_1gpu_dev_v1/codebrain/full_finetune/seed_42/metrics.json

CSBrain Unified60 Equivalence Detailed Report

Original reproduction vs patched unified60 final ablation - seed 42, 15 epochs, matched subset. Generated in a consistent format for GitHub tracking.

Verdict: CSBrain is equivalent after aligning montage, checkpoint, recipe, and effective input scale. The main fair pair is S3_original_scale_div100 vs S4_unified_bipolar_div100. S3 reaches B-Acc 0.7933 and AUROC 0.8599; S4 reaches B-Acc 0.7884 and AUROC 0.8685 (-0.49 pp B-Acc, +0.86 pp AUROC).

1. Scope and comparison rule

CSBrain required the deepest forensic audit because the initial original reproduction was not a stable reference. The split and labels were exact, so the issue was narrowed to input semantics: true bipolar reconstruction, scale, checkpoint, and training recipe.

The strict author-loader *10000 scale was retained as a control, but the main fair pair uses the strongest stable effective scale identified by smoke and final ablation.

2. Main original reproduction vs unified60 pipeline

Stage	Original CSBrain S3	Debugged unified60 S4
Data source	Original matched PKL subset	Canonical unified60 H5 matched subset
Input / montage	true 16-channel bipolar input	true 16-channel bipolar reconstruction
Shape / scale	[B,16,10,200], /100 effective scale	[B,16,10,200], /100 effective scale
Checkpoint	ON, 293 keys loaded	ON, 293 keys loaded
Loss / optimizer	BCEWithLogitsLoss / AdamW lr=1e-4 wd=0.05	same
Scheduler / best metric	Cosine per batch, eta_min 1e-6, clip 1.0, best val B-Acc	same
Evaluation	S3 B-Acc 0.7933 / AUROC 0.8599	S4 B-Acc 0.7884 / AUROC 0.8685

3. Ablation design

Pair / row	Scale	Role	Question answered
S2 original vs S6 unified	*10000	strict author-scale control	Does literal loader-style scale work with reconstructed amplitudes?
S1 original vs S5 unified	*1000	scale-sanity control	Does reduced amplified scale give stable equivalence?
S3 original vs S4 unified	/100	main debugged fair pair	Does strongest stable effective scale give equivalence?
S7 unified endpoint-current	identity	historical control only	How did old endpoint-current behavior perform?

4. Ablation results

Row	Source	Scale	Best epoch	B-Acc	AUROC	Status
S2 original scale 10000	original PKL	mul10000	5	0.6642	0.7162	weak strict-scale control
S6 unified bipolar 10000	unified H5	mul10000	14	0.7115	0.7746	unified higher but scale weak
S1 original scale 1000	original PKL	mul1000	7	0.7302	0.7639	stable control

Row	Source	Scale	Best epoch	B-Acc	AUROC	Status
S5 unified bipolar 1000	unified H5	mul1000	12	0.7220	0.7954	stable/equivalent control
S3 original div100	original PKL	div100	3	0.7933	0.8599	main original row
S4 unified div100	unified H5	div100	2	0.7884	0.8685	main unified row
S7 endpoint current	unified H5	identity	3	0.7573	0.8398	historical control only

5. Findings and settings

- The old CSBrain mismatch was caused by scale/montage/recipe alignment issues, not a unified60 preprocessing failure.
- The main S3/S4 pair differs by only -0.49 pp B-Acc, while unified60 improves AUROC by +0.86 pp.
- Strict *10000 was tested and remained weaker; this documents why the literal loader-scale row was not selected as the main fair row.
- S7 endpoint-current trains but is not fair because it is not true bipolar and has no original PKL counterpart.
- Recommended final-row wording: use S3/S4 as the main CSBrain equivalent pair, with S1/S5 and S2/S6 retained as scale controls.

6. Reproducibility notes

Item	Value
Source report	csbrain_final_ablation_seed42_report.pdf
Output root	outputs/csbrain_unified60_final_ablation_seed42_1gpu
Main fair original path	outputs/csbrain_unified60_final_ablation_seed42_1gpu/S3_original_scale_div100/metrics.json
Main fair unified path	outputs/csbrain_unified60_final_ablation_seed42_1gpu/S4_unified_bipolar_div100/metrics.json
Epochs / seed / batch	15 epochs / seed 42 / batch size 64